

Character-Based Sequence-to-Sequence Modelling for Ukrainian Grammatical Error Correction

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Abstract

Grammatical error correction (GEC) has been studied successfully for English but remains a challenging task for morphologically complex languages such as Ukrainian. In this work, we present a character-level sequence-to-sequence model based on a bidirectional LSTM encoder and an attention-based LSTM decoder for Ukrainian GEC. We train and evaluate our model on the manually annotated UA-GEC dataset, consisting of 33,735 sentences and covering grammatical, spelling, punctuation, and fluency errors. We then use a fine-tuned mBART50 baseline and compare it against our model.

1 Introduction

Grammatical Error Correction (GEC) in Natural Language Processing (NLP) performs automatic error detection and correction in written texts. The main error types are (Bryant et al., 2017):

- (a) orthographic (spelling and punctuation),
- (b) morphological (word ending or forms),
- (c) syntactic (word order or missing words),
- (d) lexical errors (wrong context or phrasing).

Interest in Ukrainian NLP has grown significantly in recent years. Despite this, Ukrainian remains a low-resource language due to the lack of annotated corpora (Pogorilyy and Kramov, 2020). In this work we use the first and currently the only dataset for Ukrainian, UA-GEC (Syvokon et al., 2023). Because of its morphological complexity (e.g. deep system of cases, verbal aspects), Ukrainian GEC is a challenging task for token-based models (Bondarenko et al., 2023). Unlike English, Ukrainian is an inflectional language. The most common errors in Ukrainian happen on the morphological level (wrong word endings and

agreement) and lexical level (calques). Plenty of Ukrainians are native speakers in both Ukrainian and Russian, which arises from a complex historical and political background. A widespread phenomenon in Ukraine is code-switching between the two languages, referred to as *surzhyk*. It occurs across multiple linguistic levels blending vocabulary, grammar, and pronunciation from both Ukrainian and Russian (Hentschel and Reuther, 2020). Due to such complexity, we use a character-based approach in our paper. We also compare the model against a fine-tuned mBART50 baseline (Tang et al., 2021), since this model achieved the best results in the UA-GEC shared task (Syvokon and Romanyshyn, 2023a).

To summarize, our contributions are as follows:

- Train a character-based encoder-decoder model on UA-GEC and evaluate it with the F0.5 metric.
- Compare our model against the fine-tuned mBART50.
- Analyze performance based on error types and discuss the pros and cons of a character-based approach for a fusional language.

2 Dataset

Ukrainian belongs to low-resource languages and UA-GEC is the first professionally annotated dataset for Ukrainian GEC. It consists of written texts, both from native and non-native speakers, and annotated by two different annotators. The annotation has the following format: {error=>edit::error_type=Tag}, where *error* is the original text, *edit* is the correction, and *Tag* denotes the error category (Syvokon et al., 2023).

Errors in UA-GEC are divided into four types, namely grammar, fluency, spelling and punctuation. There are two corpus versions in m2 format, namely *gec only* and *gec + fluency*. In this paper, we

concentrate on the latter one. `gec + fluency` is divided into train and test sets, with 31,038 and 2,697 sentences, respectively (Syvokon et al., 2023).

The `m2` format is standard in GEC and its structure is shown in Table 1. Line `S` contains a sentence and `A`-lines are annotations stating error location, its type and the corresponding correction, made by two different annotators.

Line	Content
S	She go to the store every day .
A	1 2 G/Number goes REQUIRED -NONE- 0
A	1 2 G/Number goes REQUIRED -NONE- 1

Table 1: An example of a `m2`-formatted sentence.

To use the sentences in training, they must be parsed first. UA-GEC repo provides already parsed sentences as source and target text files, so no additional parsing is required (Syvokon and Romanynshyn, 2023a).

3 Models

3.1 Character-Level Encoder-Decoder

As the model’s name suggests, it is a character-level model consisting of two parts, namely the Encoder and Decoder.

3.1.1 Character-Level Vocabulary

There are several approaches to text segmentation for `seq2seq` models: word-level, subword level (e.g. BPE or WordPiece), and character-level tokenization. A word-level model treats different inflected forms of the same word (e.g. different cases or moods) as completely different tokens, which is problematic for fusional languages as Ukrainian (Syvokon et al., 2023). BPE merges character sequences from the training data based on how often they appear together into single tokens. As a result, a model may end up cutting the Ukrainian word in the wrong place and thus deviate from morpheme boundaries (Bostrom and Durrett, 2020). Character-level models are more reliable at capturing the structure of morphologically rich languages than BPE, and can outperform subword segmentation entirely (Shapiro and Duh, 2018). They allow the model to recognize grammatical function of inflections and are better with orthographic errors (Xie et al., 2016). Given these considerations, we adopt a character-level approach for a fusional language like Ukrainian.

To build the vocabulary, we used the train dataset of both source and target sentences. As a result, we

received 322 distinct characters, of which only 70 are Cyrillic. Although the Ukrainian alphabet consists of 33 letters, the model treats their upper and lower case forms as separate symbols (66 in total), plus there are four Russian-only letters found in the data (Э, э, Ы, ё). The remaining characters that explain such a high count are: 61 Latin letters, 16 digits (including superscript) and 175 other symbols such as emojis, punctuation and other signs.

Additionally, there are four special symbols we added at the beginning of the vocabulary so that the model can indicate padding (`<PAD>`), start of the sentence (`<SOS>`), end of the sentence (`<EOS>`), and any unknown symbols (`<UNK>`). The (`<UNK>`) symbol is reserved for any characters unseen during training but present in future validation and inference steps. Thus the final vocabulary for Ukrainian GEC contains 326 symbols in total.

The model cannot work with those symbols directly, it requires a numerical input. Thus, every symbol receives its own numerical ID (see Table 2). Each ID is then mapped to a learned embedding vector required by the encoder and decoder.

Token	Index
<code><PAD></code>	0
<code><SOS></code>	1
<code><EOS></code>	2
<code><UNK></code>	3
‘a’	4
‘б’	5
‘B’	6
⋮	⋮

Table 2: Token-to-index mapping. Indices 0–3 are reserved for special tokens; character indices begin at 4.

3.1.2 Encoder

The encoder reads erroneous Ukrainian sentences, one character at a time and converts each character into a vector. Each character ID first attaches to a learned embedding vector through an embedding layer. Next, the embeddings are processed by a *single-layer bidirectional LSTM*, meaning it reads each sentence in both directions, left-to-right and right-to-left. This lets the model understand each character better, based not only on the context before the character, but also on what follows. For Ukrainian it plays an important role, since agreement and case endings often rely on the words that come after. There are two outputs that the encoder

produces: a reading of every character (used later by attention) and the final hidden and cell states of a fixed length (summary).

Forward and backward readings form two hidden states that are concatenated into one. Similarly within the cell states, their directions get concatenated and become one cell state. After concatenation the size of both states needs to be reduced to the decoder's expected size. By applying two separate *linear projections*, followed by a *tanh activation*, we receive the decoder's starting point.

3.1.3 Attention Mechanism

Without attention, the encoder must compress everything it read into a single summary vector that is handed to the decoder to generate its output. Since this vector has a fixed size, longer sentences lose some information in order to fit in. Such a squeezed vector can result in the decoder's degraded accuracy (Bahdanau et al., 2015). To solve this problem we use Bahdanau's attention mechanism that lets the decoder look back at every encoder output directly, and weight source characters to how relevant they are at that moment.

3.1.4 Decoder

The decoder's task is to write correct sentences. When it receives the encoder's summary, it starts predicting correct characters, one at a time, left-to-right. By reading the special tokens <SOS> and <EOS> the decoder knows when the sentence begins and ends. Our decoder contains an attention mechanism and thus it is no longer limited to the encoder's summary only. With attention, it builds the context vector at every single step of the decoding process. Both the context vector and the previous character's embedding are fed into the decoder's LSTM as well as the final output projection. This allows the model to look back and notice what source characters are the most relevant at the current step. The output is generated sequentially and not all at once because the context relies on the decoder's constantly updating hidden state (Bahdanau et al., 2015).

During training the decoder makes predictions (called *logits*), but they are never fed back into the model. Instead, *teacher forcing* is used, meaning the model always receives the correct previous characters. This enables a faster training process.

The inference time is where the decoder uses its own predictions and actually demonstrates what it has learned. After training, the best model is

saved and loaded again to correct new sentences at inference time.

3.1.5 Sequence-to-Sequence

Sequence-to-Sequence (Seq2Seq) is what actually brings the encoder and the decoder with attention mechanism to life. It is a wrapper that connects the two of them into one model and defines how the training and inference proceed. It also masks <PAD> tokens so that the attention mechanism ignores them.

3.2 mBART50

mBART50 is a transformer-based encoder-decoder pretrained on 50 languages, including Ukrainian. Since this model achieved the best results on the UA-GEC shared task (Syvokon and Romanyshyn, 2023b), we use it as our baseline. We fine-tune `mbart-large-50` that has about 610M parameters. Pre-trained as a denoising autoencoder, mBART50 learned how to reconstruct the original, clean text from noisy input (Tang et al., 2021). This resembles the GEC task where an error can be viewed as noise and by denoising erroneous sentences we receive the correct ones. In our case, we perform monolingual translation. Since mBART50 was created for translation tasks across multiple languages (Tang et al., 2021), we force the model at every generation step to use Ukrainian as the first token for output. Otherwise it could use any of the 50 languages it knows.

Unlike the character-level LSTM, mBART50 does not rely on a character vocabulary, since it uses its own pretrained SentencePiece subword tokenizer (Kudo and Richardson, 2018). The model has already seen plenty of Ukrainian words in pre-training, and thus, it contains representations of most Ukrainian word forms, unlike with the LSTM that starts training with zero language knowledge. By fine-tuning mBART50, the existing representations are adapted to the correction task. It is important to note that the subword tokenization may still cut morpheme boundaries since it is only frequency-based and has no knowledge of what a morpheme actually is.

4 Training LSTM vs Fine-tuning mBART50

The hardware used for both training and fine-tuning was 16 GB RTX 4060 Ti, using the `gec+fluency` training split. The hyperparameters are summarized in Table 3.

Setting	LSTM	mBART50
Parameters	~4M (est.)	~610M
Tokenization	character	SentencePiece
Optimizer	Adam	AdamW
Learning rate	1×10^{-3}	3×10^{-5}
Effective batch	48	32
Max length	800 chars	256 tokens
Precision	fp32	bf16
Best epoch	16	2
Decoding	greedy	beam (5)

Table 3: Training and fine-tuning configuration for the character-level LSTM and the fine-tuned mBART50 baseline.

4.1 Training the LSTM

Both the encoder and the decoder use a single LSTM layer with a hidden size of 256 and 64-dimensional character embeddings. We used Adam optimizer with a learning rate of 10^{-3} and a batch size of 48.

The LSTM was trained with early stopping (patience of 5 epochs) and evaluated on a held-out validation set. After each epoch we tracked the validation loss. Training halted after 21 epochs, because the last five epochs following epoch 16 showed no improvement. We restored the checkpoint with the lowest validation loss of 0.1643 at epoch 16. Character accuracy for that epoch was 0.965. This high number is explained by little character-difference between the source and the target sentences in UA-GEC.

4.2 Fine-tuning mBART50

mBART was fine-tuned only 3 epochs, with the best evaluation loss of 0.2024 at epoch 2. We perform full fine-tuning, so all ~610M parameters are updated. Full fine-tuning requires a 32-batch, storing of all activations and fp32, which is too much for a 16 GB hardware. To conduct full fine-tuning and avoid running out of memory, we used the techniques provided by HuggingFace: *gradient accumulation*, *gradient checkpointing* and *bf16 mixed precision* (Hugging Face, 2024). Activation memory depends on the batch size, so by reducing the batch size from 32 to 4 we save up a handsome amount of GB. The model processes 8 batches of 4, accumulating the gradients until the final batch, so we end up with a 32-batch. *Gradient checkpointing* enables the model to store only few activations, without intermediate ones. Instead, the latter ones

are recomputed during the backward pass. *bf16 mixed precision* shrinks some of the default 32-bit float to a 16-bit float (Hugging Face, 2024).

5 Evaluation

To evaluate the performance of the LSTM and the fine-tuned mBART50, we reuse two publicly available scripts from the UA-GEC repository, namely tokenization and evaluation (Syvokon and Romanyshyn, 2023a). The evaluation script is based on ERRANT (Bryant et al., 2017). ERRANT requires tokenized input, so each model’s output is first tokenized with Stanza. Next, ERRANT compares the originally annotated sentences from `valid.m2` with each model’s predictions and produces precision, recall and $F_{0.5}$ metrics.

There are two ways ERRANT scores a model. The main difference between them is how they evaluate the suggested fix. *Span-based detection* counts a predicted edit as correct whenever it overlaps with the actual error, regardless of the proposed correction. With *Span-based correction* the predicted edit must not only locate the error but also match the gold replacement exactly (Bryant et al., 2017). For instance, if the gold edit is гарний → гарна and the model predicts гарний → гарне, it is counted only under detection, since the location is right but the correction is wrong.

We also conduct an error-type analysis to see what kind of errors the models miss. The category labels are not produced by ERRANT, whose classifier is built for English, but taken from the gold UA-GEC annotations in `valid.m2`. For each gold edit we check whether the model produced an edit with the same span and correction and group these matches by category to calculate recall for each category.

6 Results

ERRANT evaluation on span-based correction and span-based detection is shown in Table 4. The fine-tuned mBART50 model outperforms the character-level LSTM across every metric. Both models have precision roughly twice as much as recall. This suggests that when a model proposes a correction, it is usually right, but it carefully fixes only a small part of all errors. Recall is therefore the main weakness of both models: even mBART50 reaches a detection recall of only 0.41, meaning that 59% of errors are never located. Since $F_{0.5}$ gives precision more weight than recall, it rewards this careful be-

haviour, so the scores look better than the actual coverage.

Detection scores are consistently higher than correction scores for both models, so both models are better at noticing that something is wrong than at making the exact right fix.

Model	Precision	Recall	F _{0.5}
<i>Span-based correction</i>			
LSTM (char-level)	0.4805	0.2600	0.4108
mBART50 (fine-tuned)	0.6997	0.3744	0.5961
<i>Span-based detection</i>			
LSTM (char-level)	0.5790	0.3033	0.4899
mBART50 (fine-tuned)	0.7891	0.4110	0.6665

Table 4: ERRANT evaluation on the UA-GEC validation set, comparing the character-level LSTM against a fine-tuned mBART50.

Error type	N	LSTM	mBART
Punctuation	639	0.455	0.669
Spelling	415	0.318	0.320
F/PoorFlow	138	0.029	0.127
F/Style	127	0.016	0.134
G/Case	90	0.022	0.122
F/Calque	80	0.037	0.091
G/Structure	46	0.022	0.083
F/Repetition	35	0.000	0.118
F/Collocation	21	0.048	0.174
G/Prep	18	0.000	0.185
G/Number	15	0.000	0.143
G/Gender	13	0.000	0.278
Overall	1708	0.260	0.374

Table 5: Per-category recall for the most frequent error types on the UA-GEC validation set. Overall covers all categories; only the most frequent are listed.

According to Table 5 models mostly correct local errors. The LSTM does well on punctuation (recall 0.46) and spelling (0.32), but rarely corrects grammar or fluency errors: its recall is only 0.02 for case and 0.00 for number and gender. mBART50 is better in every category but follows the same pattern, namely it is strongest on punctuation (0.67) and spelling (0.32) and weak on case (0.12) and calque (0.09). Both models therefore work mainly as punctuation and spelling correctors, while the harder morphological errors, which are characteristic of Ukrainian, remain mostly uncorrected.

This distribution mirrors the dataset itself: in UA-GEC, punctuation and spelling make up roughly 43% and 19% of all errors, respectively (Syvokon et al., 2023). As a result, the two categories both models handle best account for about 62% of all errors, which partly explains why the F_{0.5} scores look favourable despite the weak correction of morphological errors.

7 Limitations

The two models also differ in many ways at once, so the F_{0.5} gap between them cannot be traced to a single reason. mBART50 is much larger (~610M vs. ~4M parameters), pretrained on 50 languages instead of trained from scratch, uses subword instead of character tokenization, and decodes with beam search instead of greedy decoding.

All experiments ran on a single 16 GB RTX 4060 Ti. Because of this limited memory, mBART50 was fine-tuned for only two epochs with a small batch size, and the LSTM used a single, lightly tuned configuration. Neither model was trained to its full potential, and both, mBART50 in particular, would likely improve with more compute and longer training. We report a single training run per model, without multiple seeds or significance tests.

Finally, as noted in Section 6, punctuation and spelling dominate UA-GEC, so the F_{0.5} is largely driven by these frequent and comparatively easy categories. Therefore the more difficult errors are still essentially unsolved.

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